Year Up United
Data Analyst Training Academy: Introduction to Power BI

Week 9 Lab Workbook

This courseware is to be distributed to accompany instructor led training.

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Module 1

Introduction to Power BI

Exercise 1.A Getting Started with Power BI Desktop

Getting Ready

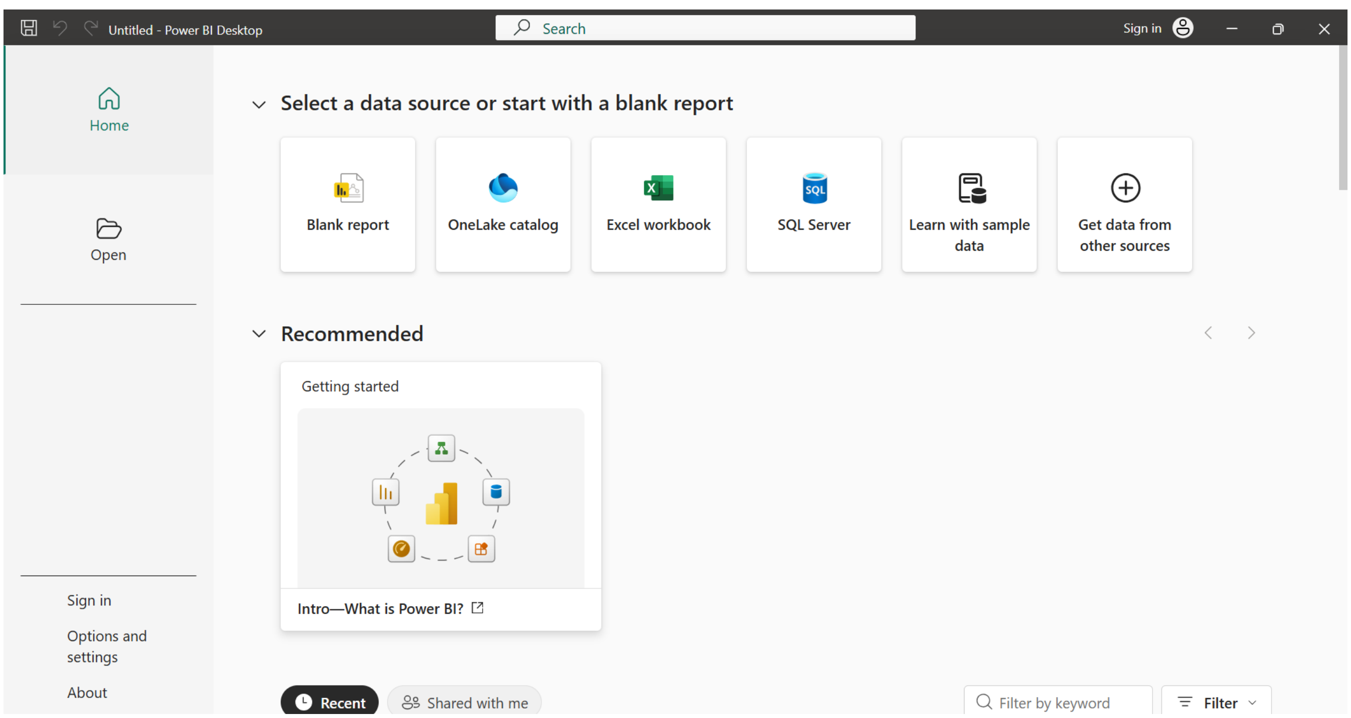
This week you will begin working with data and creating reports using Power BI in the desktop software. You will continue sharing your saved homework via GitHub.

1. On your Windows computer, open Power BI Desktop.

   * You **must** use the installed desktop software. If you do not yet have it installed, you can add it through the [Microsoft App Store]() or from the Microsoft website at https://www.microsoft.com/en-us/download/details.aspx?id=58494

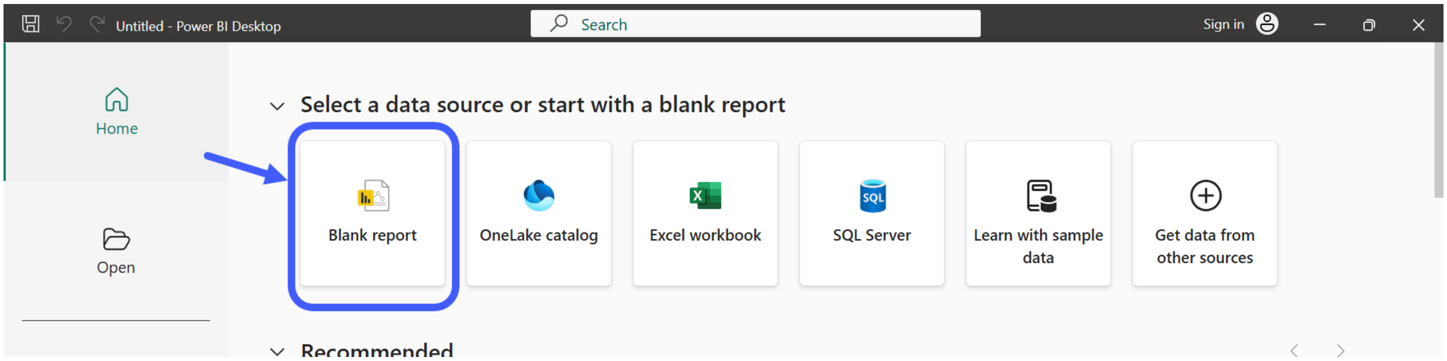
   * Power BI Desktop is not available for MacOS. You must use Windows for this software.

2. When you open Power BI, the starting page will look something like this:

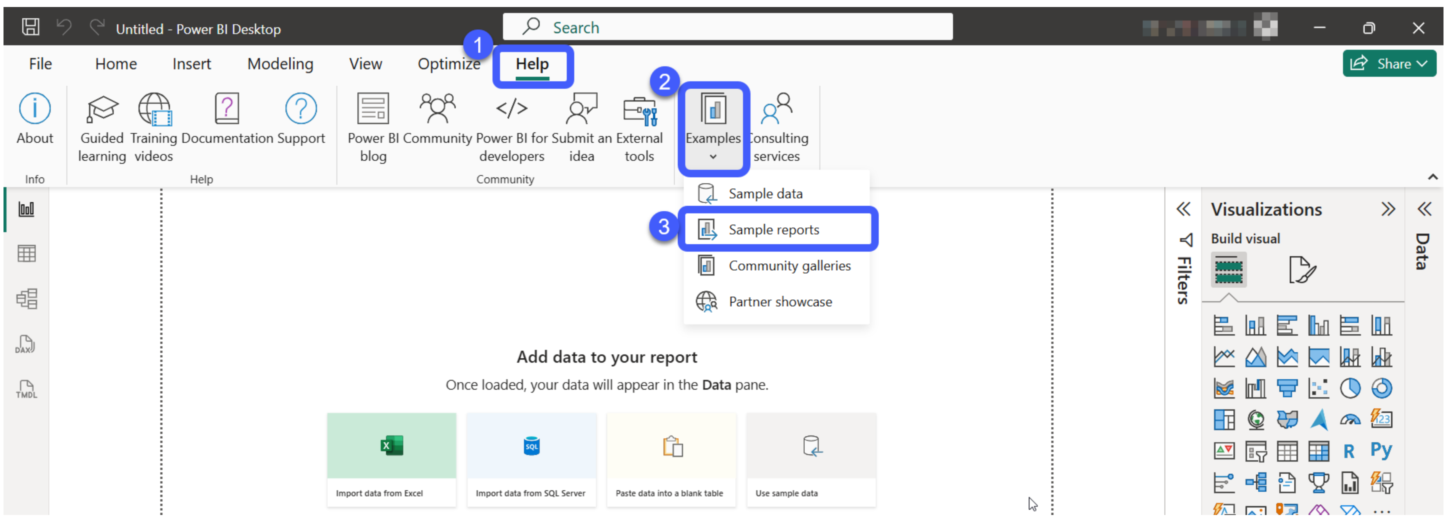


3. In the upper right corner, click **Sign in** and sign in with your yearup.org email address. Continue through any additional prompts to complete signing in with Microsoft.

4. Once you are signed in, look at the options under **Select a data source or start with a blank report** at the top of the screen, and click on **Blank report**.



5. This will open a blank report. In the ribbon at the top of the screen, click on the **Help** tab (1), then click **Examples** (2) and select **Sample reports** (3):



6. This will open to the Microsoft Learn page [What are Power BI samples](). Skim the information at the top, then scroll down to the section **Download and understand Power BI sample files** with the table of available samples.

7. Find the report named Customer Profitability sample and click the .pbix link to download a copy.

Customer Profitability sample .pbix .xlsx	Analyzes revenue, costs, and customer segments to identify high- and low-profit customers and lifetime value. Learn more about the Customer Profitability sample.

8. Proceed to Exercise 1.B.

Exercise 1.B Exploring Power BI Sample Projects

In this exercise, we will learn to navigate around the Power BI Desktop Interface by exploring a few of the freely available Power BI Desktop Sample projects provided by Microsoft at https://docs.microsoft.com/en-us/power-bi/create-reports/sample-datasets

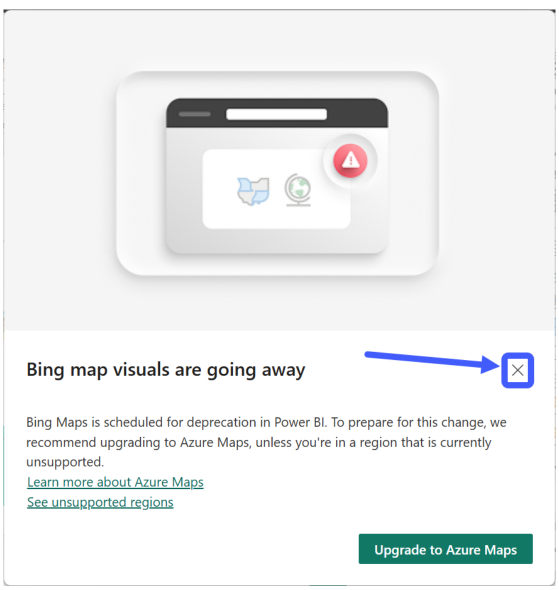
(For convenience, copies of a selection of sample reports are also saved to the course material.)

As you go through this exercise, take your time and do not rush. The goal of this exercise is to help you get oriented and see what Power BI can do. You do not need to write out answers to every question, but do take the time consider each question thoughtfully.

1. Open Power BI Desktop.

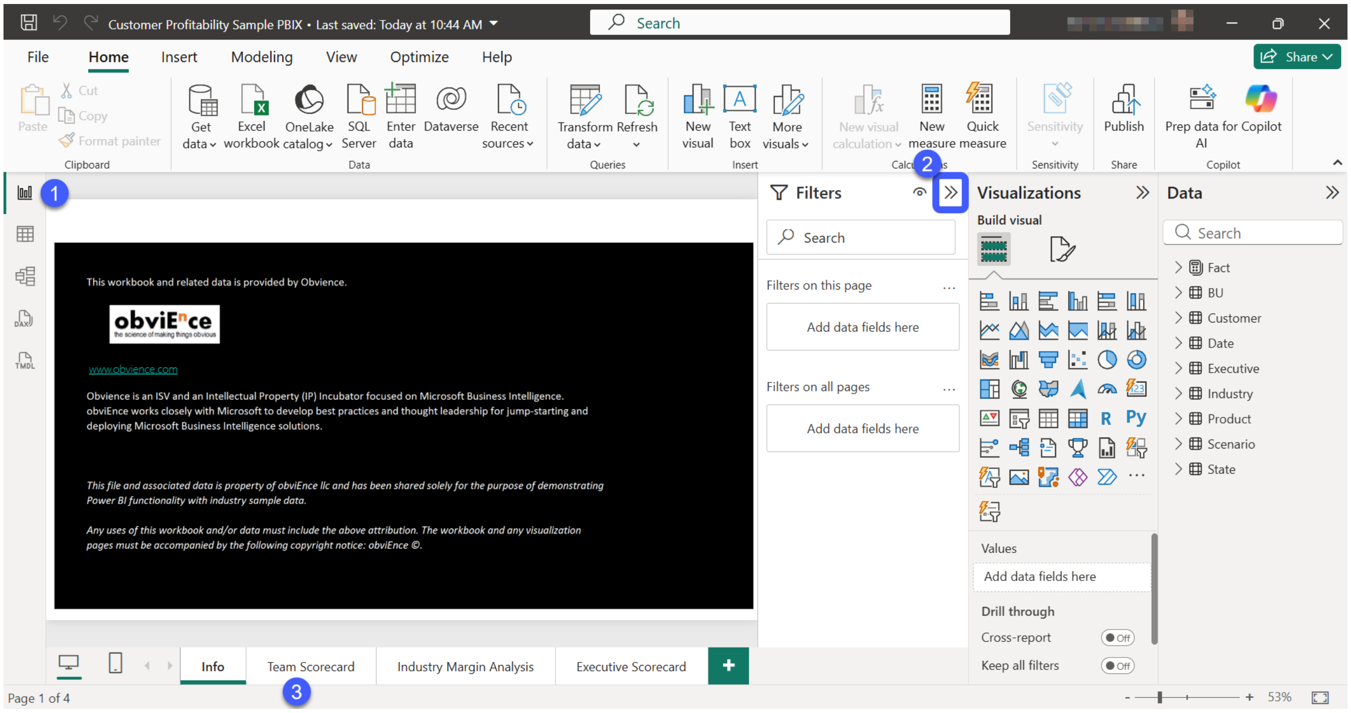
2. Open the sample file **Customer Profitability Sample PBIX.pbix**

3. As the file opens, you will get a pop-up message warning that "Bing map visuals are going away." Click the X to dismiss the pop-up message.



4. The file will open to the **Report View** (indicated by the icon on the left, see 1 below).

* If the Filters panel on the right is expanded, click the » icon to collapse it (2).
* Notice that the report has multiple tabs at the bottom. Click the tab for Team Scorecard (3).



5. On the Team Scorecard page of the report, take a few minutes to study the information displayed.

* What does this report seem to be about?

* Look at the labels for each visualization. What do you think “Revenue Var %” means? How about “RevenueTY”?

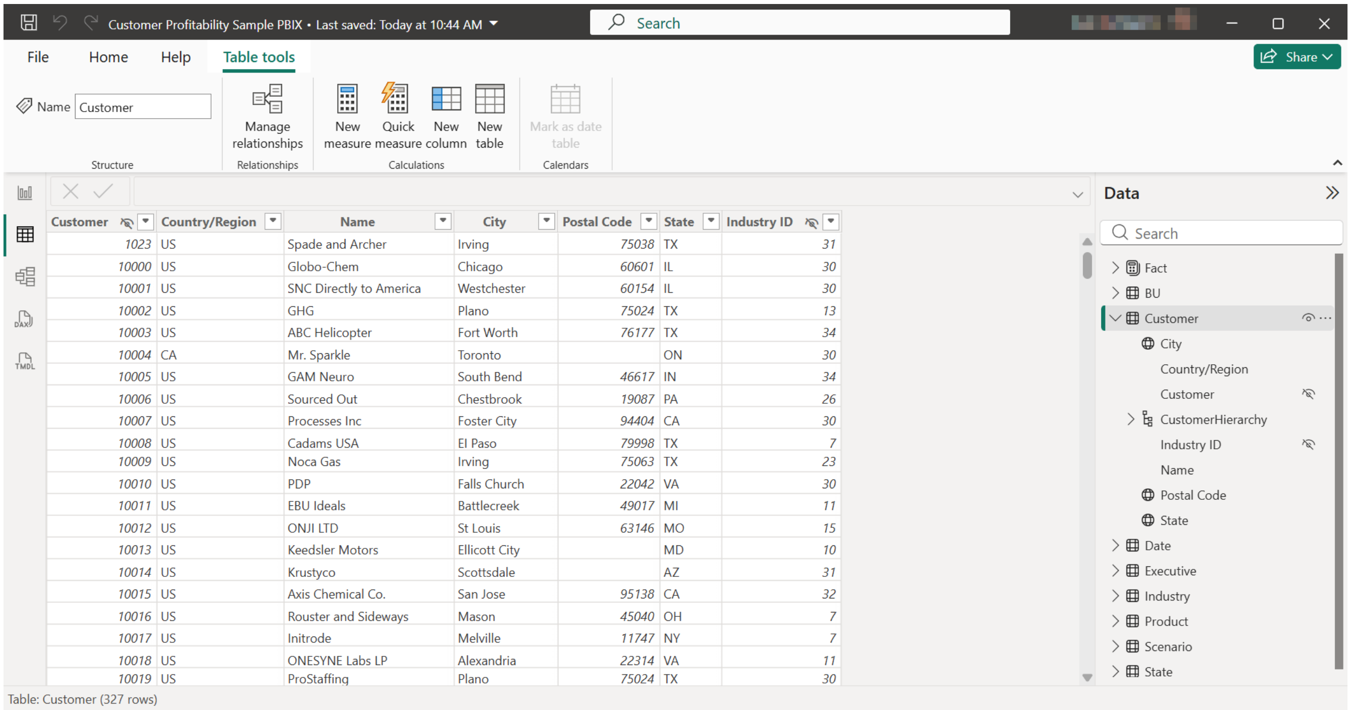
* What happens if you click on Alaska on the map visualization? (When you are ready to move on, click on Alaska again to restore the original view.)

* What happens if you click on one of the checkboxes under Name on the left? Use this feature to compare how many customers and products each team member is responsible for.

* In the visualization with all the rectangles, Total Revenue by Region, what happens when you click on the rectangle for a particular region, e.g. NORTH? (Click again to restore the original view.)

* Total Revenue by Region is currently a treemap visualization. When you click on that visualization to select it, you can also see that the treemap icon is selected in the Visualizations panel at the right. Try changing the visualization to a pie chart by clicking the icon. Which version do you prefer, the treemap or the pie chart? Why?

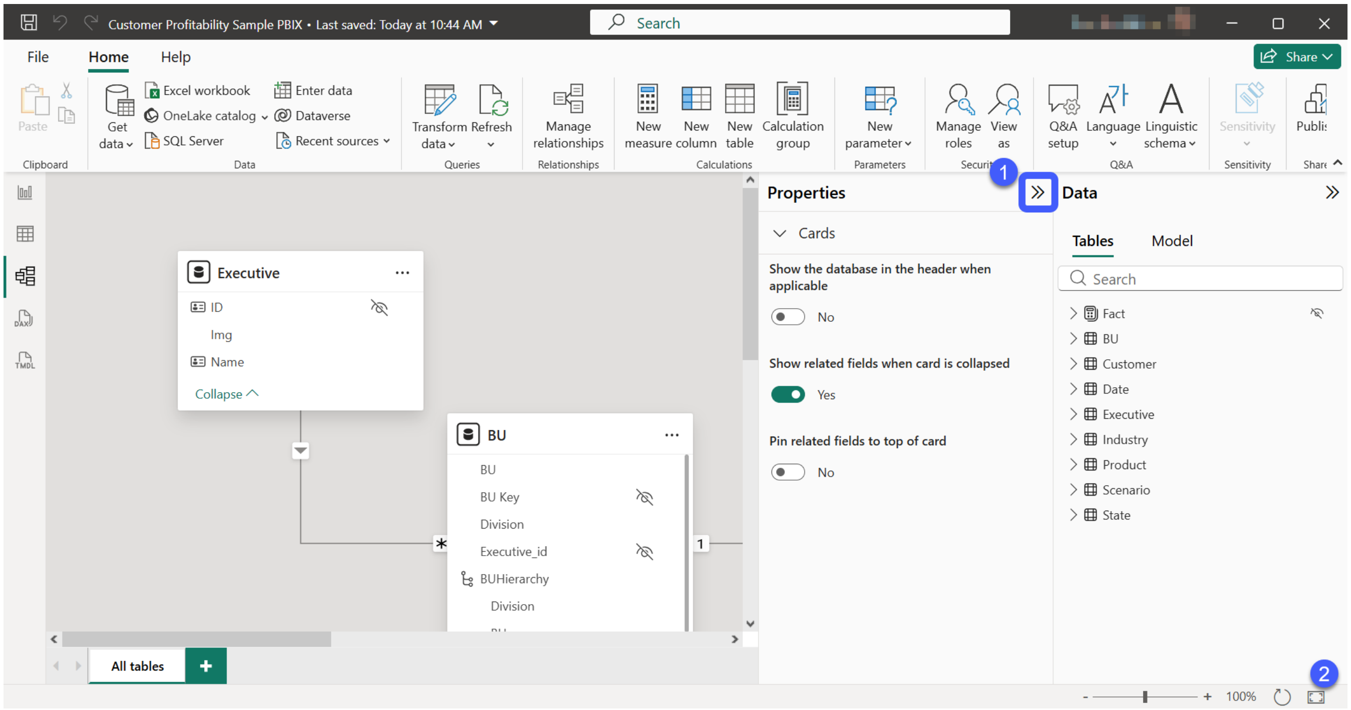
6. Navigate to the **Table View** (the icon on the left).



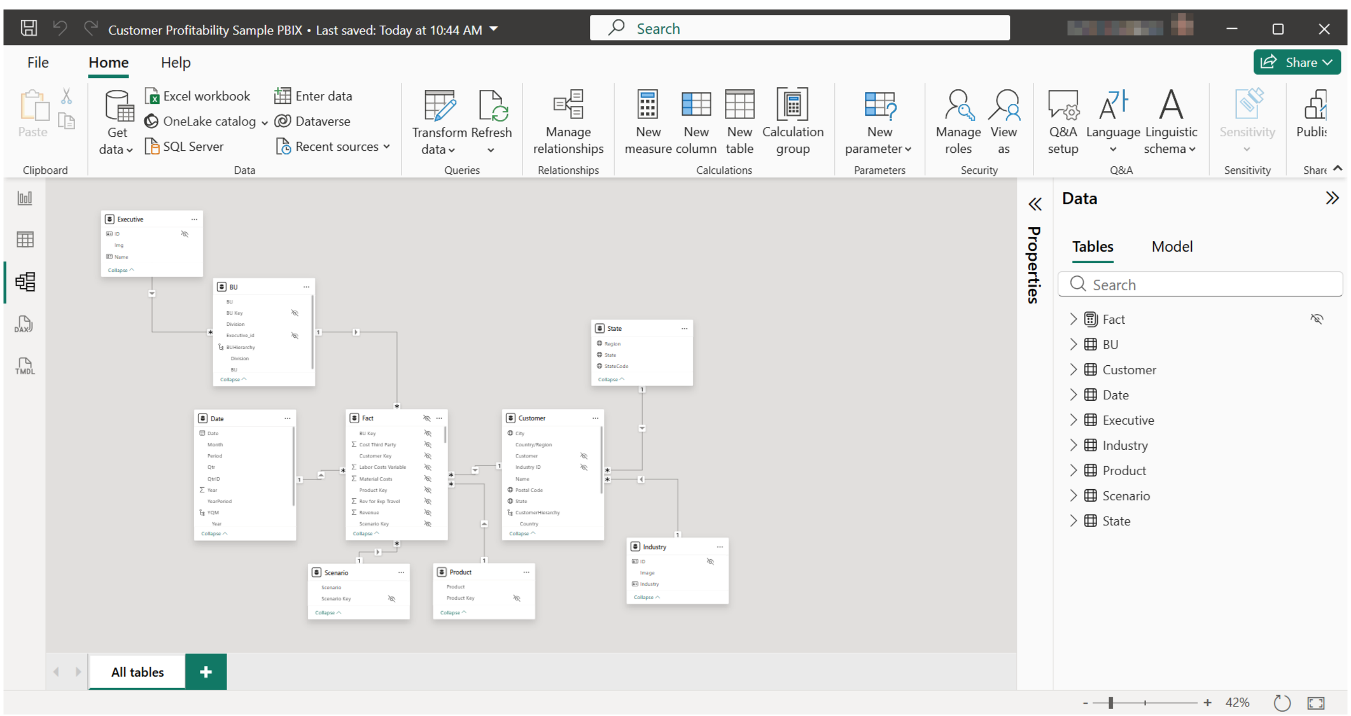
- Observe the table names in the Data pane on the right.
- Click on the **Customer** table to select it, then click the › icon to expand the table information.
- Notice the icons next to some of the fields in the table. What do you think each means?
- Notice the icon that looks like a crossed off eye next to only some of the fields. What do you think that means?
- Take a few minutes to explore other tables and fields.

7. Navigate to the **Models View** (the icon on the left).

- You might need to collapse the Properties panel (1 below) and click the fit-to-screen icon (2) to get a better view of the model.



Updated view after using fit-to-screen:



8. Observe how the various tables are related and consider the following questions. Do not worry if you don't have answers yet, we will be exploring each of these subjects later.

* What table is at the center?

- For tables branching out from the center, do they connect to other tables?
- What kind of shape does this layout seem to make?
- What kind of information shows up in the center table, compared to the outer tables?

9. Navigate back to the **Report View** (the icon on the left).
   - Take a few minutes to the other report pages in the **Report View**, using the tabs at the bottom to navigate between pages. What data points are being visualized in each of the pages? What happens when you click on various components of the visuals?

10. Select 2 more sample files (either downloading directly from the Microsoft Learn sample site or from the course materials) and explore each of them in a similar way. Spend at least 5-10 minutes on each sample. The purpose of this exercise is to create questions in your head, and help you begin to think about ways that Power BI can be used to analyze data.
    - Take your time and feed your creative mind with ideas and questions!

11. When you are finished, make a post to the class Teams channel:
    - List the names of the reports you reviewed, and provide a short (one sentence) description of each report.
    - In your post, write **at least one question** that you have about **each** of these reports. This can be a question about what something in the report means or what that information would be used for; something that is not clear from the report itself; something that you think would be interesting to investigate further; or any other question that you have.

Module 2

Data Modeling with Power BI

Exercise 2.A Exploring the Northwind Database

Introduction

The Northwind database is a sample database provided by Microsoft. The database was originally created for Microsoft SQL Server 2000. You previously installed a MySQL version of this sample database in Week 2 of the course.

This database contains data from a fictional company named **Northwind Traders**. This is a database that has been highly normalized. Normalization has separated data into many different tables and eliminated as much duplicated data as possible.

In real-world projects, it makes sense to take a little bit of time to research the source data that we'll be working with. As we do this research, keeping records of our findings will help keep us organized and effective. **Reviewing and understanding your data is a critical step in any data analysis project!**

This exercise will give you the opportunity to re-familiarize yourself with the Northwind database and make some notes that will be used in later exercises to create a useful Power BI Model. Save your work as directed to **DataAnalytics** > **week-09**

Exploring Northwind

1. Open **MySQL Workbench**.

2. Expand the schema for the **`northwind`** database that you previously created. (If you do not have northwind currently in your list of available schemas, add it using the instructions from the end of the Week 2 Lab Workbook.)

3. Expand **tables**.

   * Take a moment to carefully consider each table. What do you believe a record in the table represents?

4. Below is a list with the total number of records in each table. Since it's a sample database, there are not a lot of records.

Table Name	Row Count
`Categories`	8

`CustomerCustomerDemo`	0
`CustomerDemographics`	0
`Customers`	93
`Employees`	9
`EmployeeTerritories`	49
`Order Details`	2155
`Orders`	830
`Products`	77
`Region`	4
`Shippers`	3
`Suppliers`	29
`Territories`	53

Note the ratio of records between tables. Is this ratio consistent with what you thought the data represented?

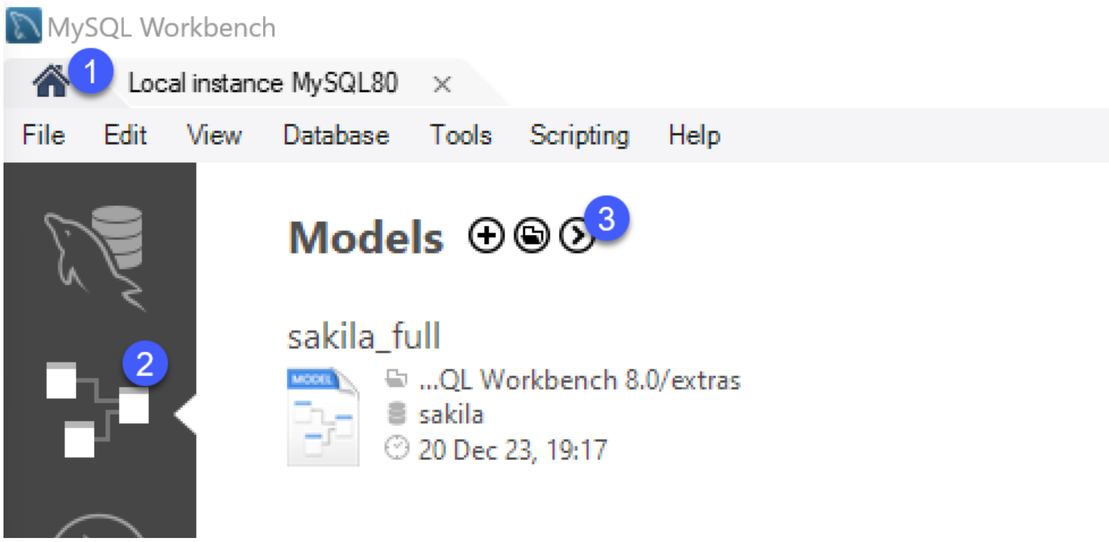
5. Note that the **CustomerCustomerDemo** and **CustomerDemographics** tables do not contain any records. For this reason, we will not be using these two tables in our analysis. Right-click and choose Drop Table to remove these from the database.

6. Use the Information panel in Workbench to review basic data about each table:
   * What is the primary key of the table?
   * What are the parent tables of this table? (i.e. What tables do any foreign keys reference?)

7. Expand the **Columns** folder under the table in the schema panel. Jot down your notes to the below questions in a text or markdown file saved as **northwind_review.txt** or **northwind_review.md** (you can use Notepad on your computer for this). Note: it may be helpful to copy and paste the list of questions into your text document first, then proceed with answering each question.
   * What does a value in this column represent? What values might you see here?
   * Is this column a part of the primary key to this table?
   * Is this column a part of a foreign key that points to a record in another table?
   * Would this column be valuable to bring into our Power BI Model? Yes, or no? Why?

* Do you believe this column is appropriately named for Data Analysis purposes?
* If not, what might be a more appropriate name?
* What might be the data type and format for this column in a Power BI Model?
* Can you think of any calculations where this column data might be used?

8. Go to the Home tab 1 in MySQL Workbench, then to Models 2 using the icon on the left. Then click on the > icon 3 and select **Create EER Model from Database.**

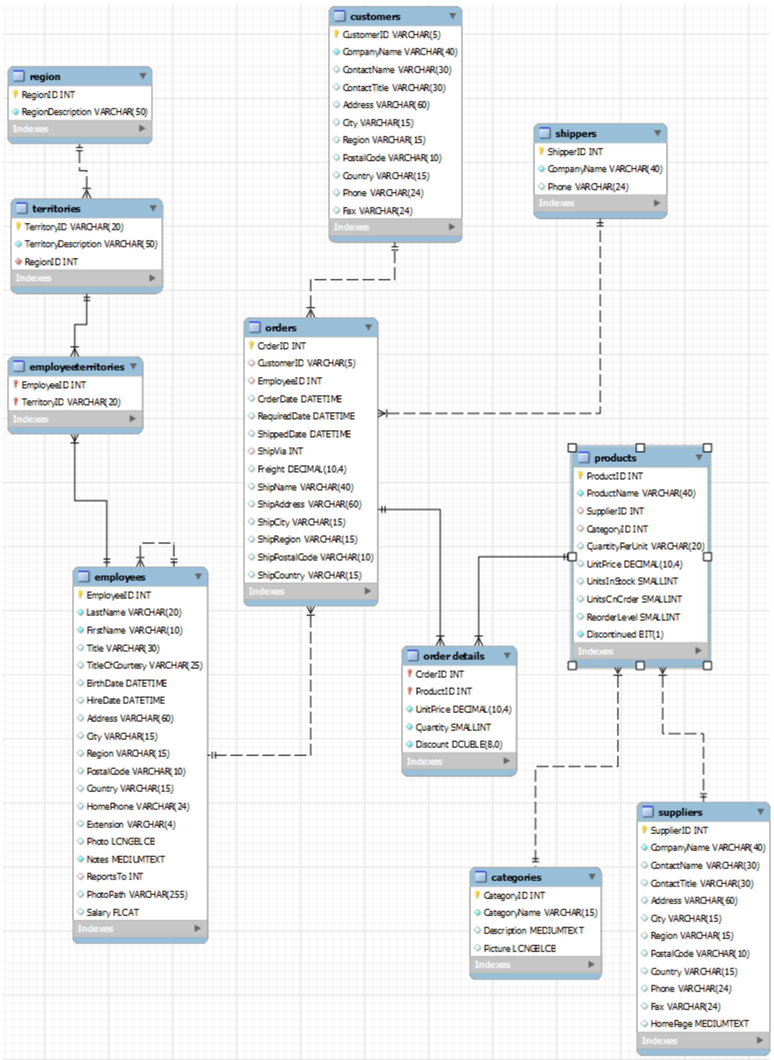


9. In the "Reverse Engineer Database" wizard:

* Leave the default settings for "Set Parameters for Connecting to a DBMS" and click **Next.**
* For "Connect to DBMS and Fetch Information" click **Next.**
* For "Select Schemas to Reverse Engineer," check the box next to **northwind**. Then click **Next.**
* Wait for a moment for "Retrieve and Reverse Engineer Schema Objects" to finish running, then click **Next.**
* For "Select Objects to Reverse Engineer" leave the check next to "Import MySQL Table Objects." Uncheck other the other two Import options. At the bottom, leave the check next to "Place imported objects on a diagram" and click **Execute.**
* Wait for "Reverse Engineering Progress" to complete, then click **Next.**
* On the final "Reverse Engineering Results" screen, click **Finish.**

10. Review the resulting diagram, and take a few minutes to arrange the tables visually in a way that makes sense to you.

    * As you click and drag tables, you will notice that they are somewhat tangled up. Try to move everything around to avoid any relationship lines overlapping or crossing over other tables.
    * Delete stray values that are not connected to any table (if any).
    * You may also wish to zoom in or out using the Zoom options at the upper left to get a better view of the diagram.



11. Use the File menu to export your model as a PNG file with the name **northwind_eer.png** and save it to your week-09 folder.

Make sure to commit and push your work to GitHub!

Exercise 2.B Continuing Northwind Exploration

Planning our transformations: Keep it or drop it?

1. The next phase of this lab asks you to consider what elements from the Northwind database to include in your Power BI project, and what elements can be excluded. To start, open the Excel file **NorthwindTablesAndColumns.xlsx** from the course material and save it to your week-09 folder.

2. Review the following guidelines for columns to include or exclude:

   * Any Primary Key columns will need to stay. Mark those “keep.”
   * Any Foreign Key columns that point to a table in the model will need to be used to define a relationship between tables in the model. Keep those.
   * Any column that represents a name may be useful as we’re likely to want to aggregate on that and see the value in reports. Mark those “keep.”
   * Any column that is a longer descriptive column like Description is probably not going to be useful for analysis in Power BI. Mark those for removal.
   * Any column that represents binary data, like a photo, will not be used in this project, and can be removed.
   * When it comes to geography, we will keep columns of higher granularity like Region, Country, Postal Code, or Zip Code. But we do not need to keep columns that store a street address like Address because we will not be aggregating at that granularity of geography.
   * Columns that contain dates that we would want to report on, such as a Ship Date or Order Date are important, keep those.
   * Columns that contain Dates like Birth Date or a Hire Date might be useful as we can compute an age or list employees based on the duration they have been at the company, keep those.
   * Columns that contain dates that are there for auditing purposes such as Last Updated On are not useful to us, remove those.
   * Columns that represent phone numbers, email addresses, and other ancillary contact information are not useful to us, remove those.

3. Pass through the spreadsheet once for all Northwind columns, and fill out the following spreadsheet columns using the guidelines listed above:
   * Primary Key?
   * Foreign Key?
   * Keep this column?

Planning for Analysis: Changing Column Names

4. Creating intuitive names will make downstream modeling and visualizing easier, so let us take a moment to consider new names for each of our columns.

5. Here are a few guidelines to consider when naming things:
   * AddSpacesBetweenWordsSoThatLabelsDoNotLookLikeThis -- although this rule would *not* apply to Key columns, such as **ProductID**, as these are not typically displayed directly in reports.
   * Give most columns a name that is unique across all tables in the model to remove ambiguity.
     - If the **Product** table has a column named **Name**, and the **Category** table has a column named **Name**, and you bring those two columns together in a report, you'll have Name and Name. That's confusing.
     - At first thought you might name these **Product Name** and **Category Name** ... but you can further simplify these to be **Product** and **Category**, as these are the primary descriptors for these tables, and they make natural labels.
   * For columns like **Country**, provide context to the name as relevant. For example, is it **Customer Country** or **Store Country** or **Supplier Country**? This may feel redundant because nearly every column ends up getting prefixed with its table name, but in reports you often only see the column name by itself.
   * Add the word **Amount** to numeric values that will be aggregated and represent money. This will help to add clarity with columns like **Discount Amount** and **Discount Percent**.

6. Using the above guidelines, fill out the "New Name?" column in the spreadsheet to suggest new names for each column as appropriate.

Measures & Dimensions

7. Pass through the spreadsheet one more time and identify candidates for:
   * **Measures** (numeric values that would be aggregated)
   * **Dimensional Attributes** (descriptive text that could be used to group aggregated values)
     * Note, it is ok if you are unsure about these. Just give it your best guess!

8. Pass through the spreadsheet one more time, this time identifying if it will make sense to apply **formatting** to a numeric column, such as marking it as a currency or percentage.

9. Pass through the spreadsheet one more time, this time brainstorming for **calculations** that a column might later become a part of (such as using sum, count, average, or other aggregation functions). Be creative here!

10. Write down any notes or concerns that you might have about your understanding of a column that you may want to further research in the “Notes:” column.

11. Save your changes to **NorthwindTablesAndColumns.xlsx**, then commit and push to week-09.

Module 3

Power Query GUI Experience

Exercise 3.A Northwind Modeling – Power Query

Introduction

In this exercise, we will be importing data from the Northwind Database into our Power BI Model, using **Power Query**.

You will notice that we won't need to write any code, as the GUI enables us to import and perform most common transformations.

For guidance, we will be using the spreadsheet that you created earlier in Exercise 2.B. We had to put a lot of thinking and creative work into making that spreadsheet, but now that those decisions are made, we have a clear list of technical needs that we can apply in Power BI.

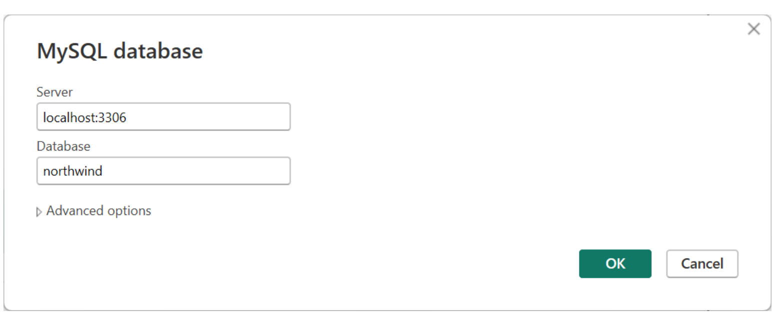
Creating our Project and Importing Data

1. Open a new blank Power BI Project.

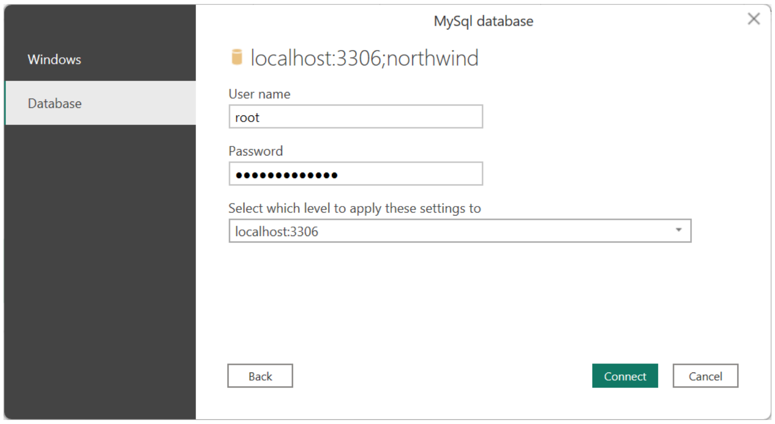
2. In the Home tab of the Power BI ribbon, Choose **Get data → More → Database** then from the list choose **MySQL database**. Click Connect.

   * **Note**: You may get a message "This connector requires one or more additional components to be installed before it can be used." If so, click **Learn more**, then download and install MySQL Connector. (No need to log in, just click "No thanks, just start my download.") Then close Power BI, reopen, and try the Get Data steps again.

3. In the **MySQL database** dialog, enter the name of the local server where you have been working with MySQL Workbench (should be **localhost:3306** but check your Workbench settings if needed). For Database, enter **northwind**, then select **OK**.

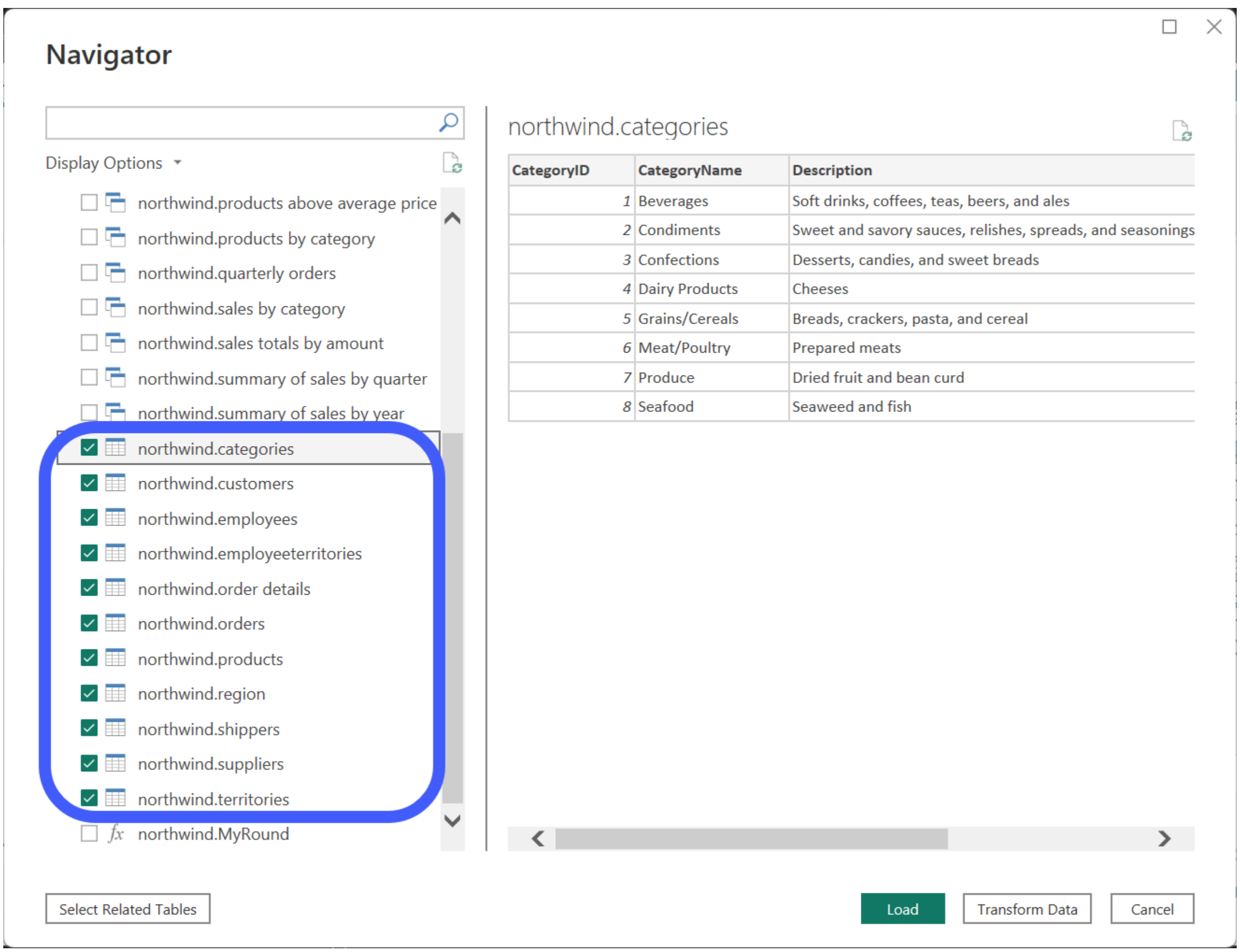


* If prompted, provide login information to connect to the server: On the **Database** tab of the pop-up dialog, enter the user name **root** and the password you created when you set up MySQL Workbench.



Make sure you use the Database tab on the left (<u>not</u> Windows).

4. In the Navigator pop-up dialog, check each of the selected tables from the image below (you will need to scroll down to find them):



5. Click **Transform Data** to open Power Query.
   (Note: if you jumped ahead and already clicked Load instead, in Power BI use the Transform Data button in the ribbon to open Power Query.)

6. Go to File > Save As, and save your project to week-09 repository as:
   NorthwindModeling_Step01.pbix

   * You will get a pop-up message informing you that you have un-applied changes in your project and asking if you would like to apply changes before saving. For purposes of these exercises, it is not important which option you choose, but it will save you a few extra seconds to choose Apply Later.

Changing Table Names

7. Rename each of the **queries** (tables) as follows, and pay attention to the plurality of each new name! We're giving **Data tables** plural names that describe the recorded event, and **Lookup tables** non-plural names that describe the dimension / attributes they represent (except where representing a many-to-many relationship).

   Quick tip: You can double-click on the name of the table in the Queries panel to make it editable, or hit the F2 key on your keyboard. When you are done updating the name, hit the return key to commit the change. You can also use the arrow keys to navigate up and down among queries.

Old Name	New Name
northwind categories	Category
northwind customers	Customer
northwind employees	Employee
northwind employeeterritories	Employee Territories
northwind order details	Sales
northwind orders	Order
northwind products	Product
northwind region	Region
northwind shippers	Shipper
northwind suppliers	Supplier
northwind territories	Territory

8. Once you have made all column name changes, save your work as:
   NorthwindModeling_Step02.pbix

Removing Unnecessary Columns

9. Open your previously-saved version of **NorthwindTablesAndColumns.xlsx.**

10. Review the **Keep it?** column of your spreadsheet. Your columns to delete should match the following list:

table_name	column_name	Keep it?
Category	Description	No
Category	Picture	No
Customer	Address	No
Customer	Phone	No
Customer	Fax	No
Employee	Address	No
Employee	HomePhone	No
Employee	Extension	No
Employee	Photo	No
Employee	Notes	No
Employee	PhotoPath	No
Order	ShipAddress	No
Shipper	Phone	No
Supplier	Address	No
Supplier	Phone	No
Supplier	Fax	No
Supplier	HomePage	No

11. In Power Query, go through each table/query and **remove** all the columns marked in your spreadsheet as columns we should not keep.

   * **Note**: When looking at the column lists, you may notice additional columns at the end of each query that represent database relationships (e.g. a column in Category for "northwind.products"). **Do not delete these!** We'll look at them later.

12. Save your work as:
    NorthwindModeling_Step03.pbix

Renaming Columns

13. Review the **New Name?** column of your spreadsheet. In Power Query, go through each table/query and rename columns as needed. Your new column names should match the following list:

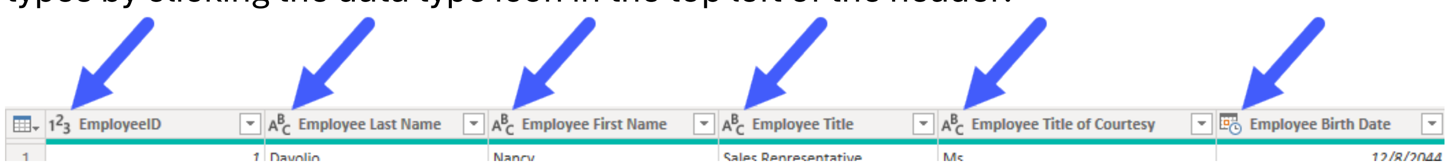
table_name	column_name	New Name?
Category	CategoryName	Category
Customer	CompanyName	Customer
Customer	ContactName	Customer Contact
Customer	ContactTitle	Customer Contact Title
Customer	City	Customer City
Customer	Region	Customer Region
Customer	PostalCode	Customer Postal Code
Customer	Country	Customer Country
Employee	LastName	Employee Last Name
Employee	FirstName	Employee First Name
Employee	Title	Employee Title
Employee	TitleOfCourtesy	Employee Title Of Courtesy
Employee	BirthDate	Employee Birth Date
Employee	HireDate	Employee Hire Date
Employee	City	Employee City
Employee	Region	Employee Region
Employee	PostalCode	Employee Postal Code
Employee	Country	Employee Country
Employee	ReportsTo	EmployeeReportsToEmployeeId
Sales	UnitPrice	Unit Price
Sales	Quantity	Sales Quantity
Sales	Discount	Sales Discount Percent
Order	OrderDate	Order Date
Order	RequiredDate	Order Required Date
Order	ShippedDate	Order Shipped Date
Order	ShipVia	ShipperID
Order	Freight	Order Freight Amount
Order	ShipName	Order Ship Name
Order	ShipCity	Order Ship City
Order	ShipRegion	Order Ship Region
Order	ShipPostalCode	Order Ship Postal Code
Order	ShipCountry	Order Ship Country
Product	ProductName	Product
Product	UnitPrice	Product Current Unit Price
Product	UnitsInStock	Product Units In Stock

Product	UnitsOnOrder	Product Units On Order
Product	ReorderLevel	Product Reorder Level
Product	Discontinued	Product Discontinued
Region	RegionDescription	Region
Supplier	CompanyName	Supplier
Supplier	ContactName	Supplier Contact
Supplier	ContactTitle	Supplier Contact Title
Supplier	City	Supplier City
Supplier	Region	Supplier Region
Supplier	PostalCode	Supplier Postal Code
Supplier	Country	Supplier Country
Territory	TerritoryDescription	Territory

14. Save your work as:
    NorthwindModeling_Step04.pbix

Assigning Column Types

15. Systematically go through the columns for each query to review and update data types by clicking the data type icon in the top left of the header:



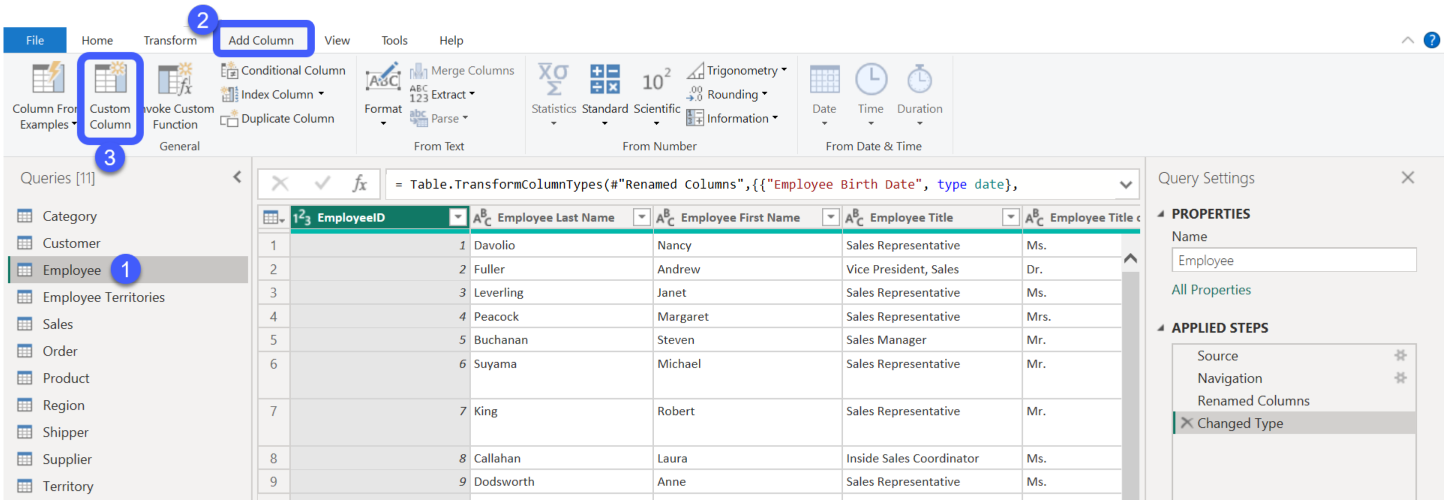
* On the **Employee** and **Order** tables, change all Datetime columns to use **Date** instead
* On the **Sales** table, **Discount Percent** should be a **Percentage**
* On the **Sales** table, **Sales Quantity** should be a **Whole Number**
* Any currencies should be **Fixed Decimal ($)**
  - on the **Employee** table, Salary
  - on the **Sales** table, Unit Price
  - on the **Order** table, Order Freight Amount
  - on the **Product** table, Product Current Unit Price

16. Save your work as:
    NorthwindModeling_Step05.pbix

Exercise 3.B Adding Custom Columns & Combining Tables

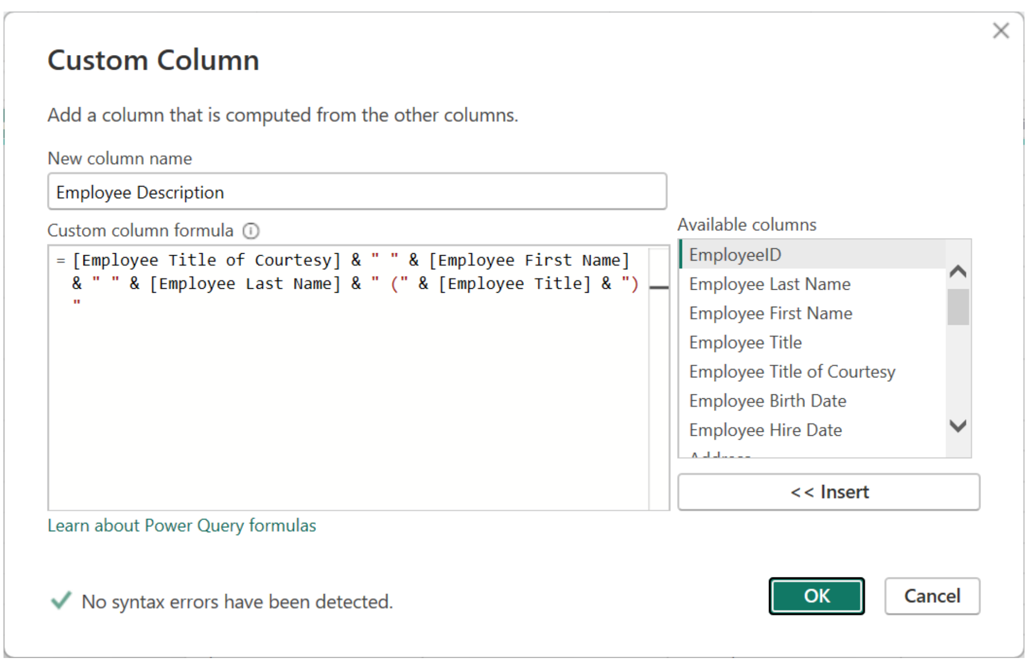
Adding Custom Columns in Power Query

1. From the Queries panel, open the **Employee** table (1). In the ribbon, click on the Add Column tab (2), then click Custom Column (3).



2. In the Custom Column window that pops up, name the new column **Employee Description**, and copy and paste (or re-type) the below expression into the **Custom column formula** box:

```
[Employee Title of Courtesy] & " " & [Employee First Name] &
" " & [Employee Last Name] & " (" & [Employee Title] & ")"
```



* Take a moment review the new column that was added, and compare it to the expression you used above. Consider how the expression logic worked to create the new Employee Description column values.

3. Add another custom column to the Employee table named **Employee**, using the following expression:
   `[Employee Last Name] & ", " & [Employee First Name]`

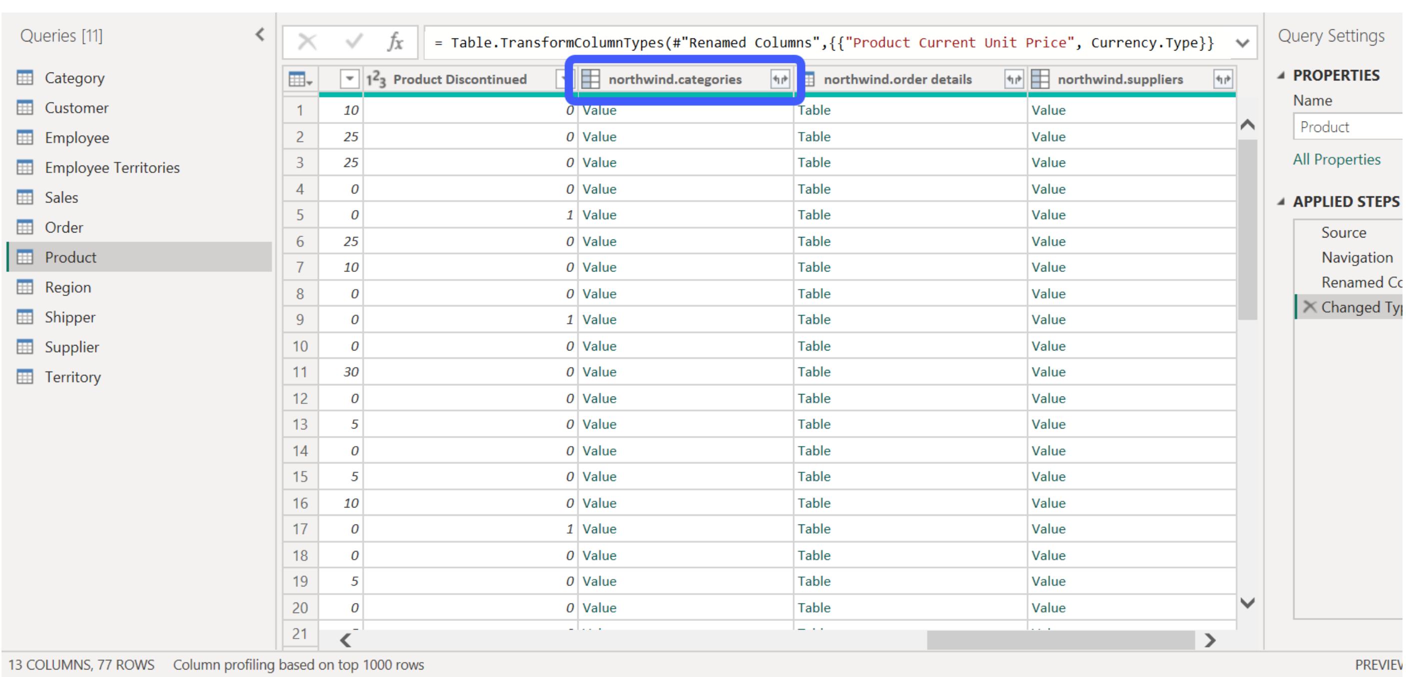
4. Switch over to the **Sales** table. Add a custom column to the Sales table named **Discount Amount**, using the following expression:
   `[Sales Quantity] * [Unit Price] * [Sales Discount Percent]`

5. Set the new **Discount Amount** column to use a **Fixed Decimal ($)** data type.

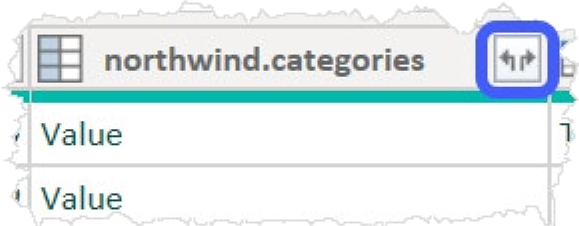
6. Add another custom column to the Sales table named **Sales Amount**, using the following expression, then set this new column to the **Fixed Decimal ($)** data type:
   `[Sales Quantity] * [Unit Price] * ( 1 - [Sales Discount Percent] )`

Using Foreign Key Relationships to Combine Tables

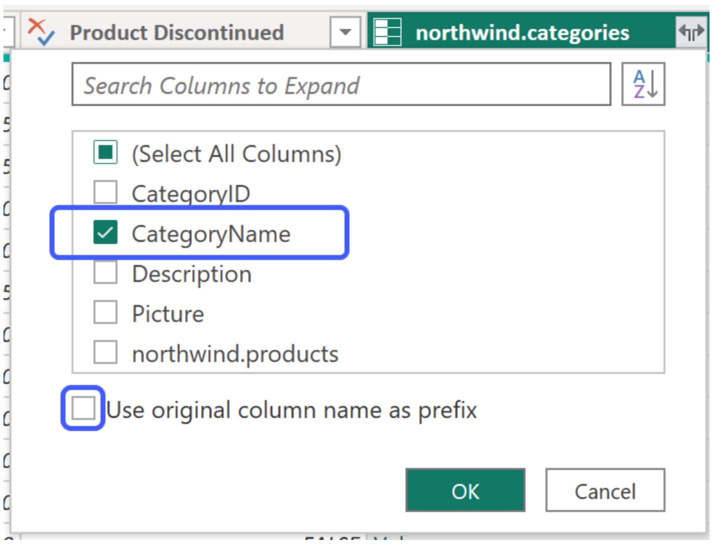
7. Open the **Product** table, and scroll to the right to find the **northwind.categories** foreign key field.



8. Click the expand button at the top right corner of the column.

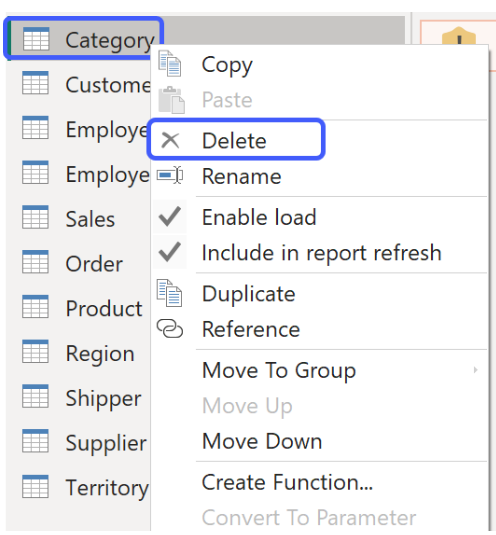


9. From the menu that appears, uncheck the box next to Select All Columns, then re-check the box just for the **CategoryName** column. Uncheck "Use original column name as prefix" so it is not selected. Then click OK.



* Rename the new column to **Category**.

* Now you can **delete** the separate Category table from the model. In the Queries panel, right-click on Category, then select **Delete**.

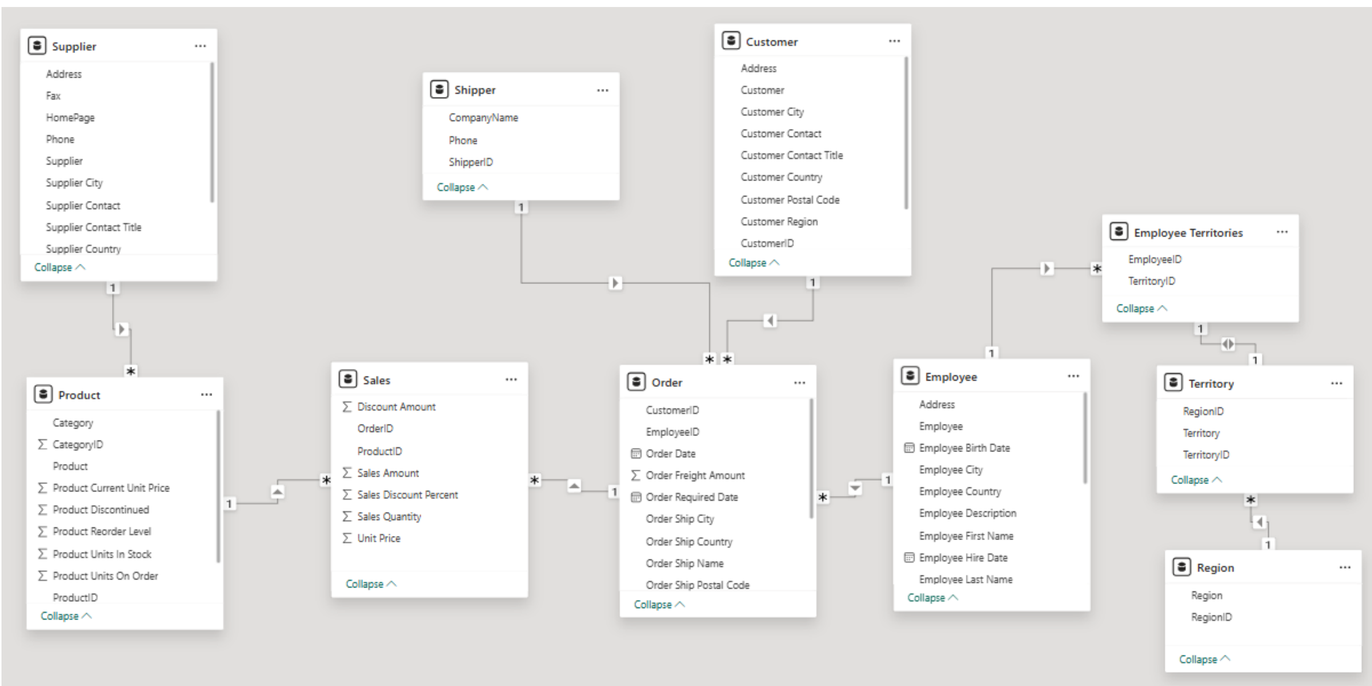


10. Save your work as:
    NorthwindModeling_Step06.pbix

Modeling and Visualizing Our New Data

11. Click **Close & Apply** to close Power Query, apply the changes to the Power BI model, and return to Power BI Desktop.

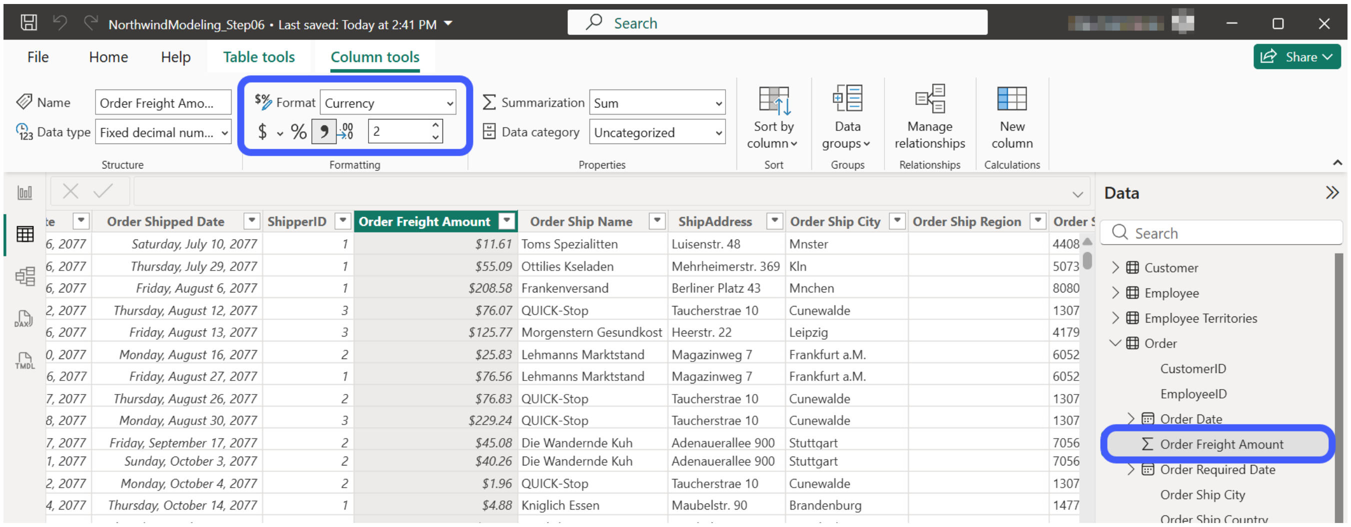
12. Go to the **Model view**, and click and drag the imported Northwind tables to arrange them in a snowflake-like model with Order and Sales toward the center. Notice that relationships between tables have been automatically detected.



13. Still in the Model view, click on the icon to next to any **key** columns (primary keys or foreign keys) to hide them from the Report view. (Icon will change to for hidden items.)

14. Use the **Table view** to review data for each table. Check out the column **formatting** options in the ribbon, and set formatting as appropriate for columns in your data.

    * For monetary values, even though you set those data types to Fixed Decimal ($) in Power Query, you will also likely need to format them as **Currency** in the Table view (see screenshot below).

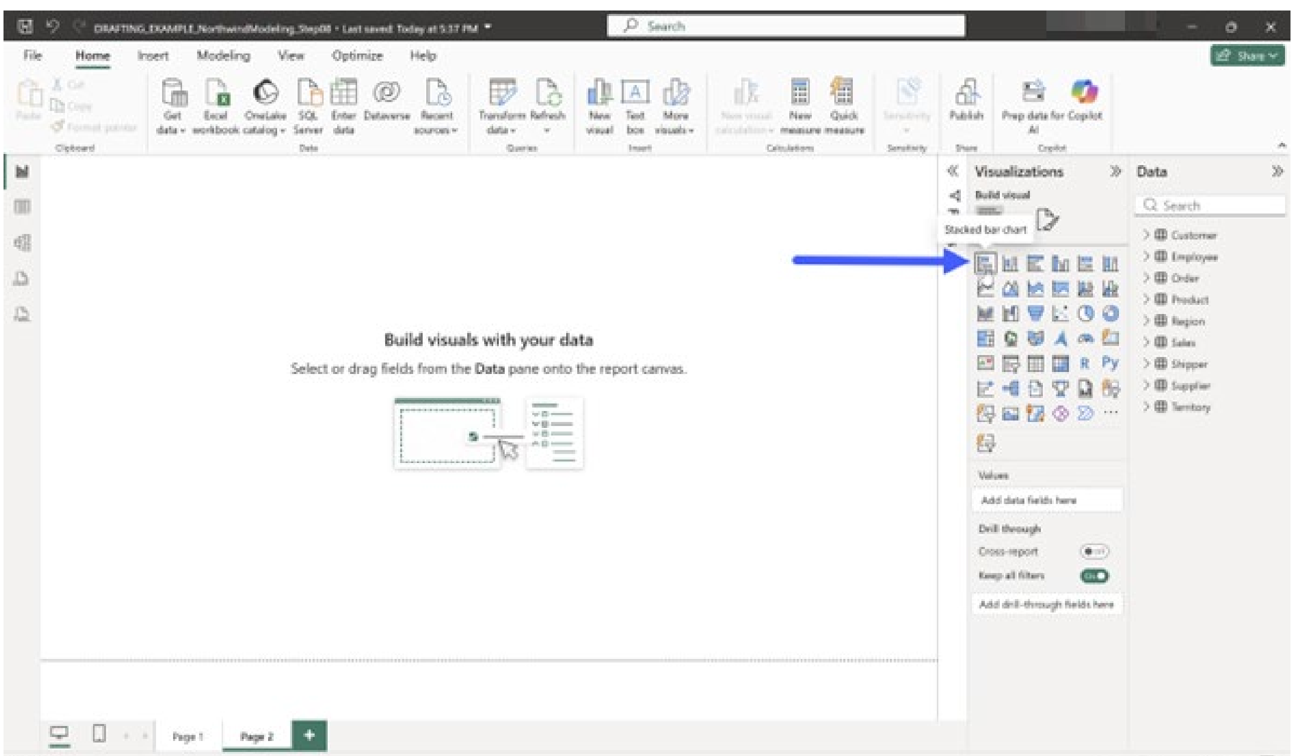


15. Still in the Table view, use the icon to hide any tables that do not contain useful columns for creating visualizations to exclude them from the Report view, such as Address fields, Fax, Phone, Photo, or similar. (Don't hide City or Country columns.)

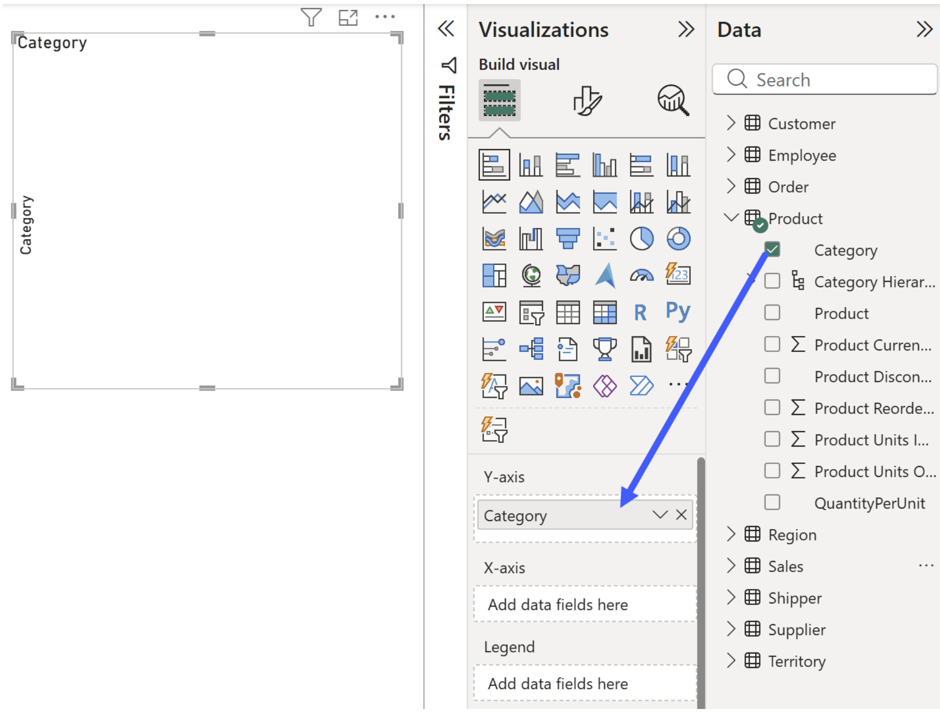
16. Go to the **Report view**. Start creating some basic visuals using fields available in your data model, following the steps below.

17. Add a Bar Chart:

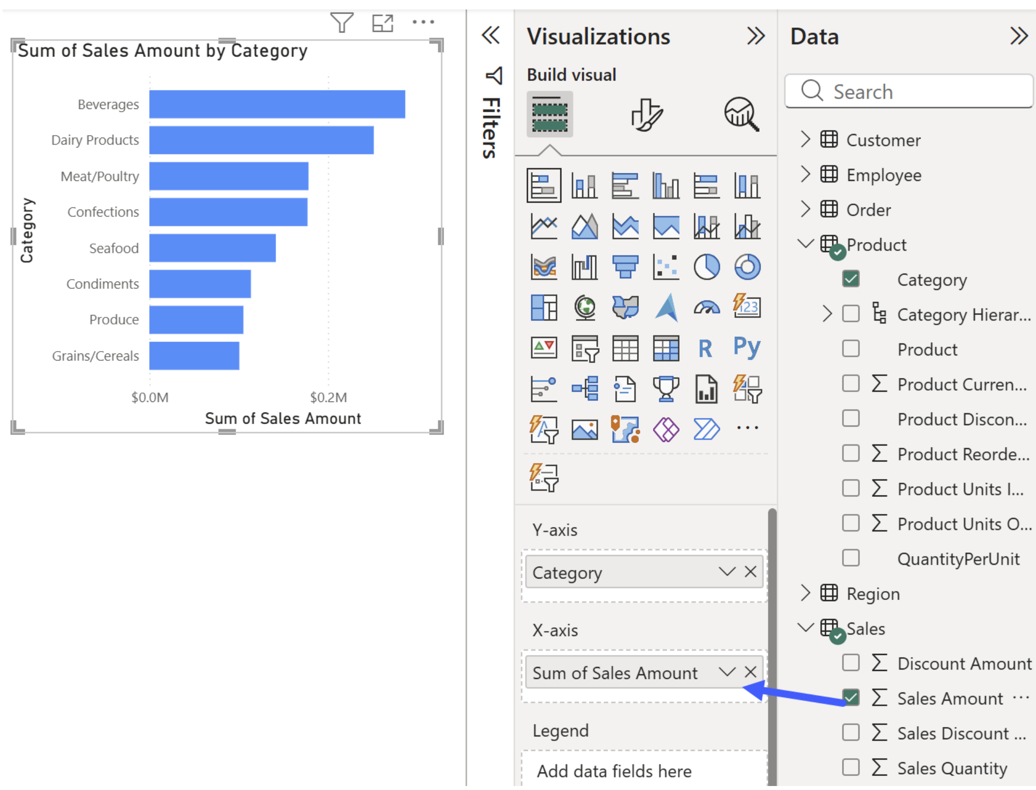
    * In the Visualizations panel on the right, click on the first bar chart icon to add a new **bar chart** to the blank report page:



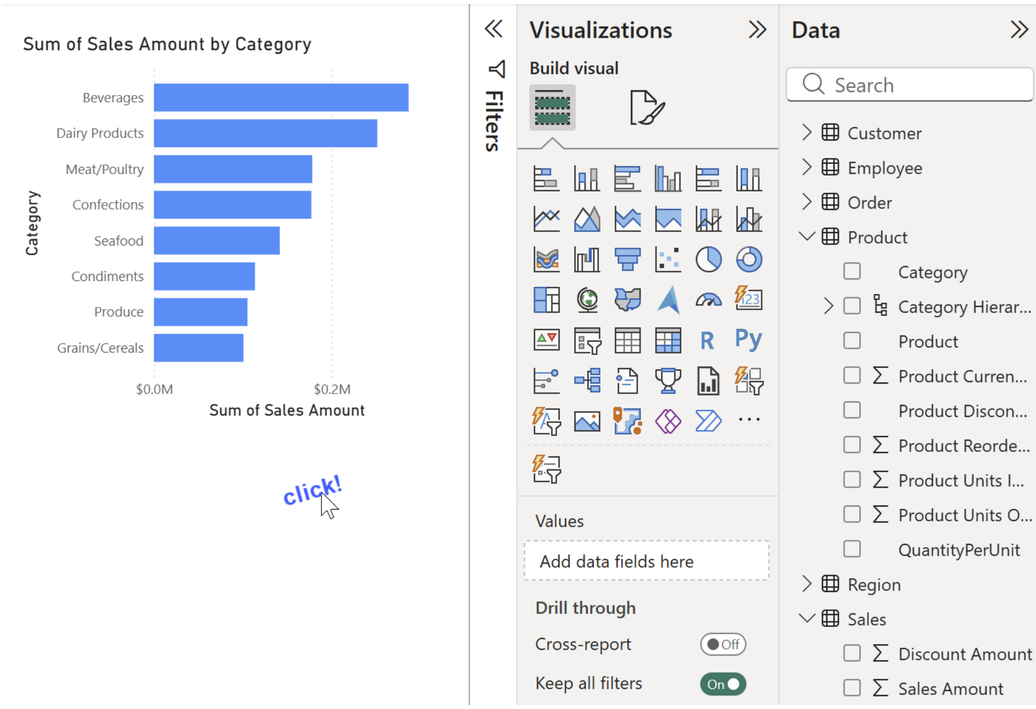
* In the Data panel, expand the Product table. Click the checkbox next to Category to add the Category field to the Y-axis of the bar chart.



* Next expand the Sales table. Click the checkbox next to the Sales Amount field to add it to the X-axis of the bar chart. When you have both Category and Sales Amount added to the chart, it should display bars for each category of items sold.

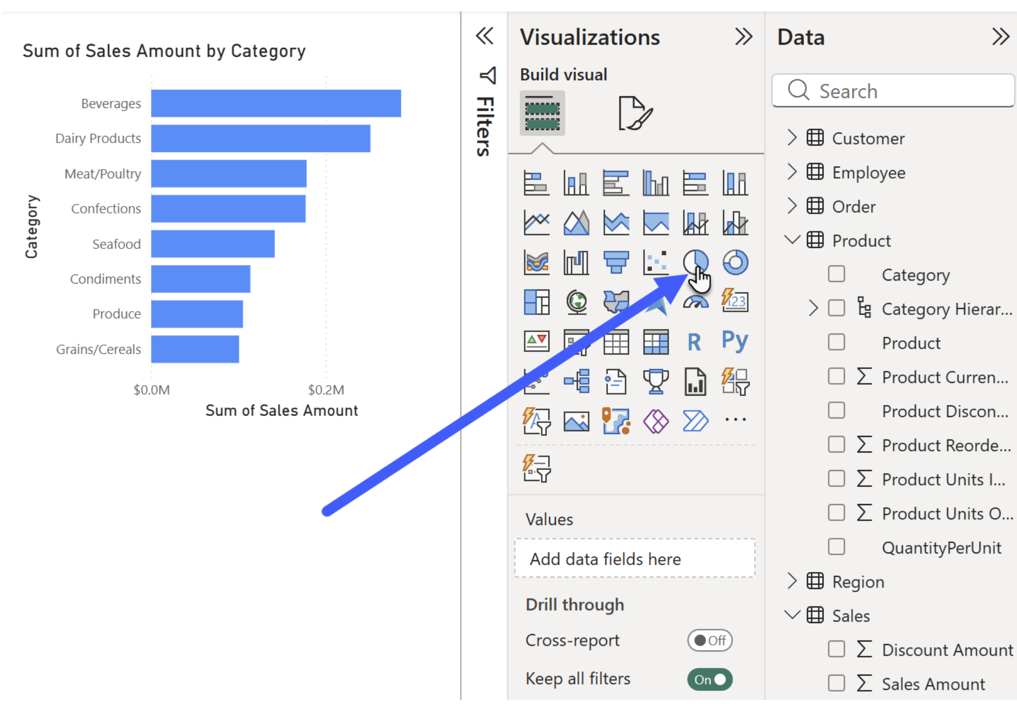


* When you are done, click in the background of the report page so that the bar chart is no longer selected.

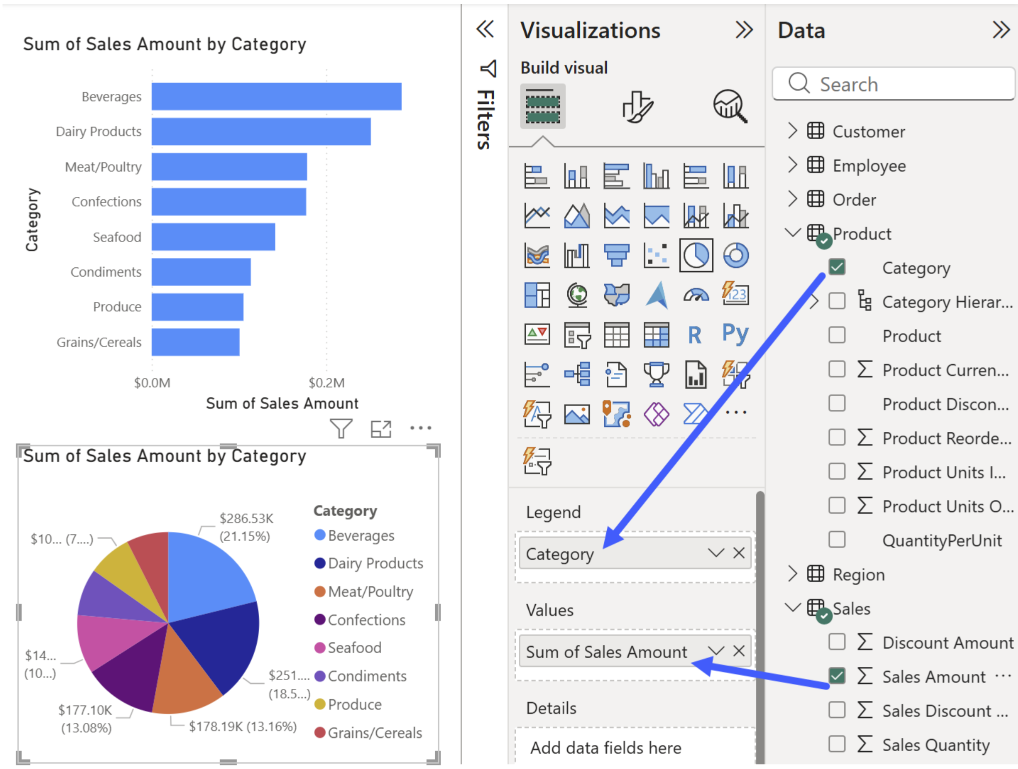


18. Add a Pie Chart:

* Make sure the bar chart is no longer actively selected (again, click in the report background to deselect).
* In the Visualizations panel, click on the **pie chart** icon to add a new pie chart.

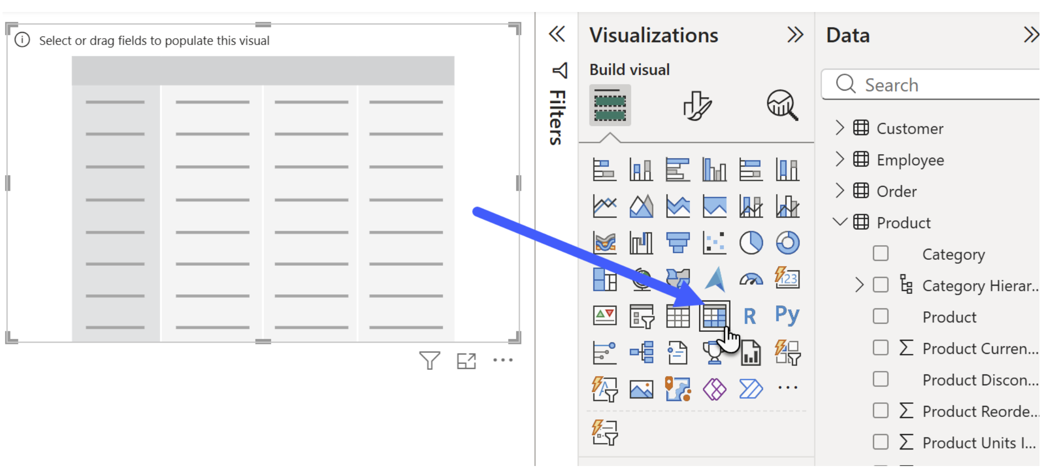


* Using the Data panel, from the Product table add the Category field to the Legend. From Sales add the Sales Amount to Values.

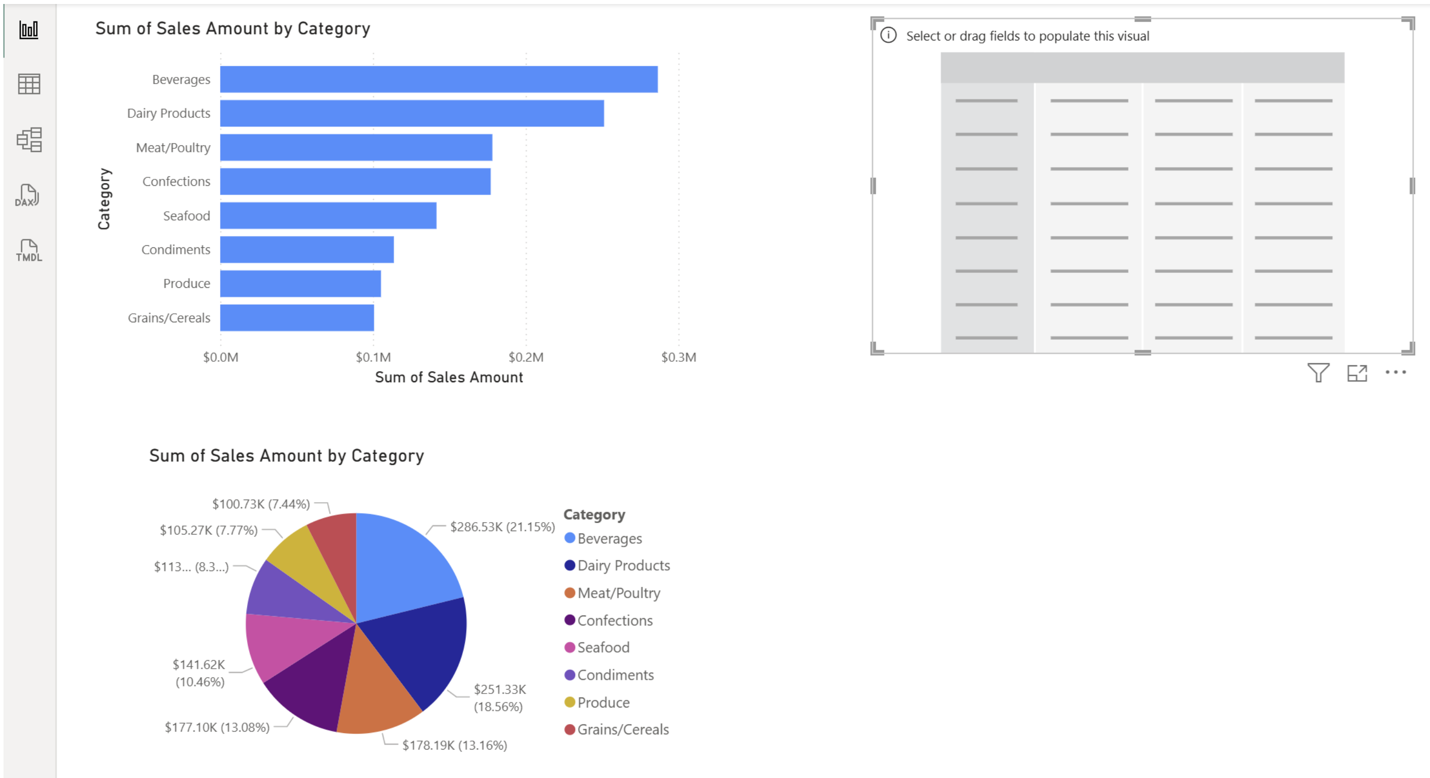


19. Add a Matrix:

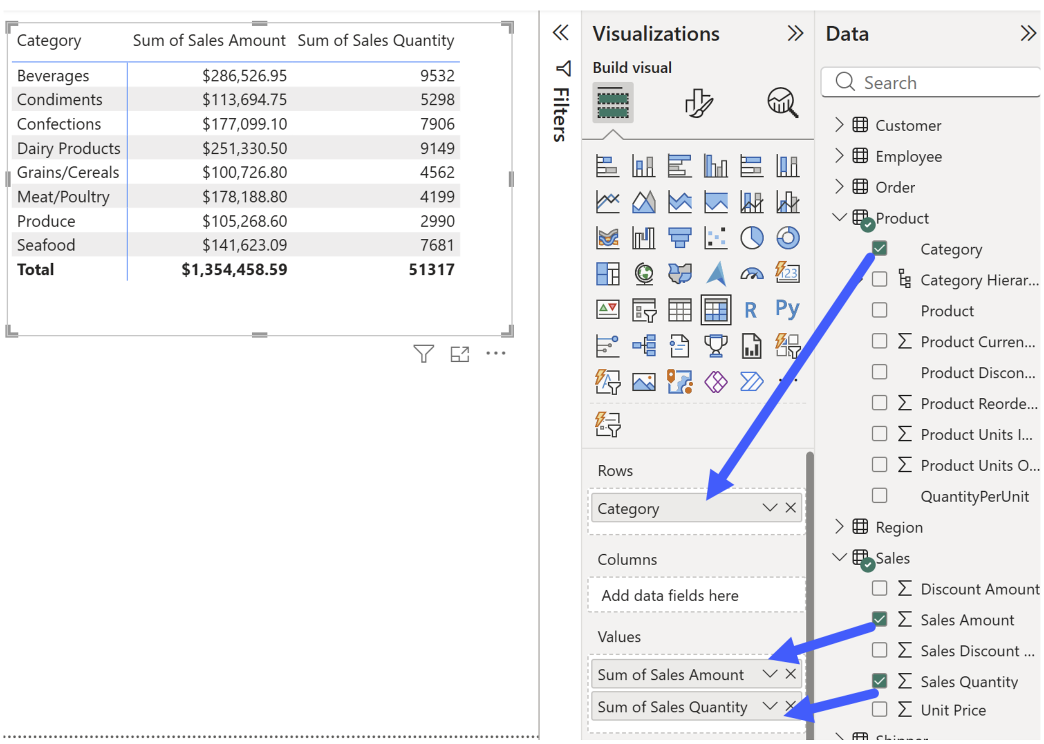
* Again click in the background of the report page so that pie bar chart is no longer selected. Then in the Visualizations panel, click the icon to add a new **matrix**.



* At this point you may want to drag your visualizations around to resize and rearrange them on the page, like this:

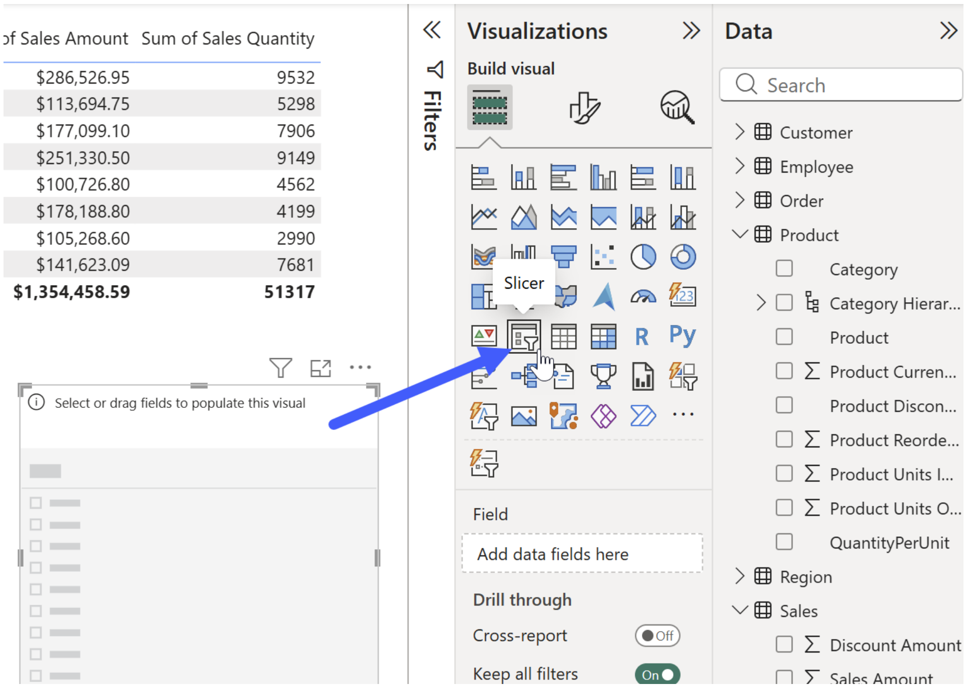


* With your matrix visualization selected, click to add Product > Category to Rows, and from Sales add both Sales Amount and Sales Quantity to Values:

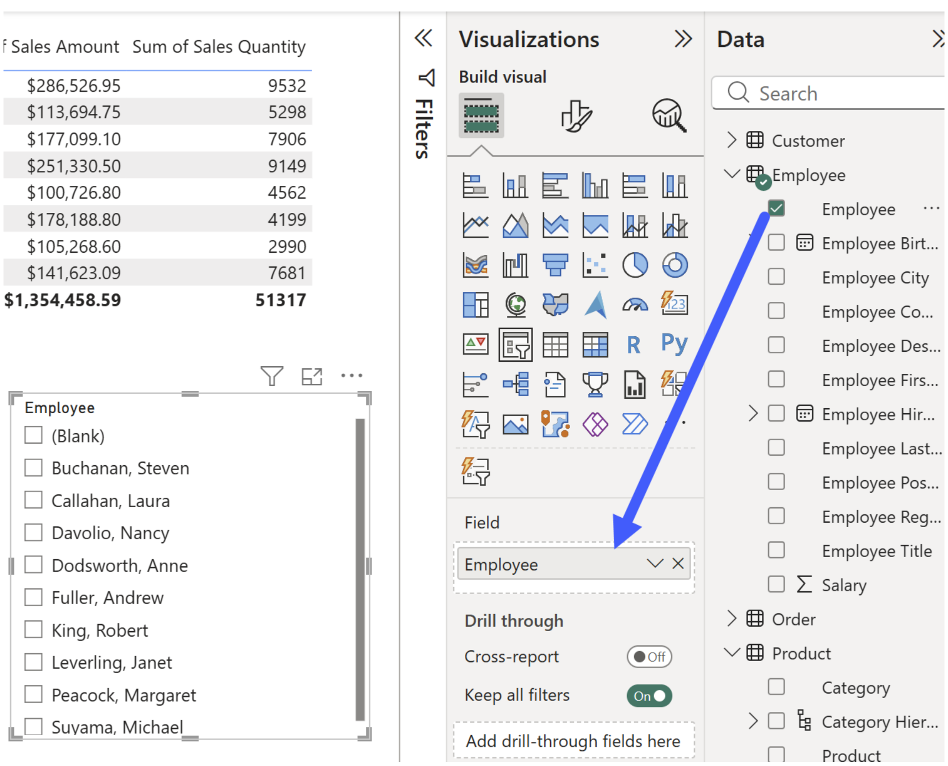


20. Add a Slicer:

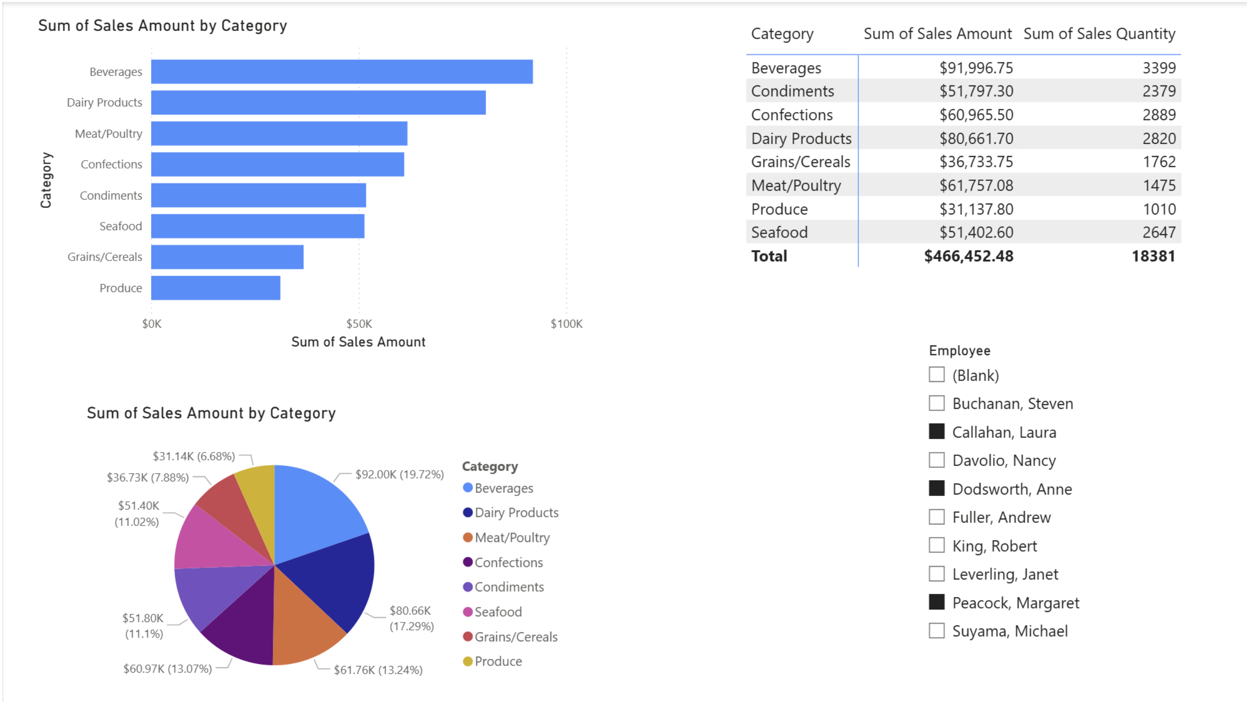
* Again click in the background of the report page so that matrix chart is no longer selected. Then in the Visualizations panel, click the icon to add a new **slicer**.



* In the Data panel, expand the Employee table. Then click the checkbox next to the Employee field to add it to the Field well. This will add checkboxes for all employee names to the slicer.



* Try checking different boxes in the slicer to see how it affects the report page. To select multiple checkboxes, you can hold down the ctrl key while clicking.



Category	Sum of Sales Amount	Sum of Sales Quantity
Beverages	$91,996.75	3399
Condiments	$51,797.30	2379
Confections	$60,965.50	2889
Dairy Products	$80,661.70	2820
Grains/Cereals	$36,733.75	1762
Meat/Poultry	$61,757.08	1475
Produce	$31,137.80	1010
Seafood	$51,402.60	2647
Total	**$466,452.48**	**18381**

21. When you are done adding visualizations, save your work as: **NorthwindModeling_Step07.pbix**

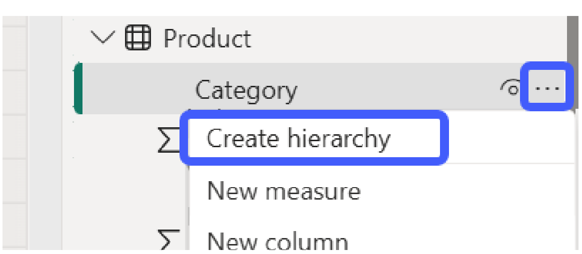
Relationships & Hierarchies

22. Review current table relationships in the **Model view**

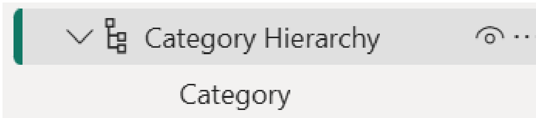
* How can you tell which relationships are one-to-many / many-to-one? Does the model contain any many-to-many relationships? Spend some time exploring the Model view to figure out how you can determine this.

23. Go to the **Table view.**

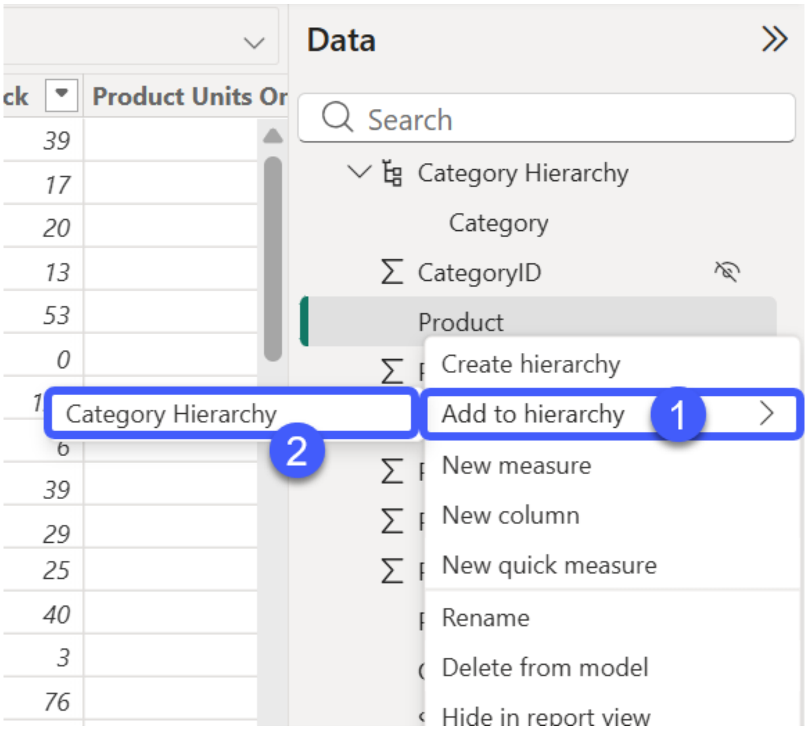
24. Expand the Product table. Select the **Category** column, then click the ellipses icon (**…**) and choose **Create hierarchy.**



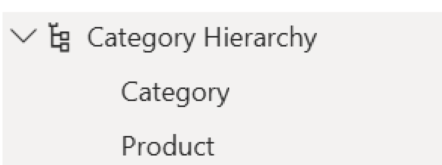
* Expand the new **Category Hierarchy** to see what it looks like. It will contain a single item, Category.



* Select the **Product** column, then click the **...** menu and choose **Add to hierarchy**(1). Then click to add this field to the Category Hierarchy (2).



* The Category Hierarchy should update to show both items, Category and Product.



25. Return to the **Report view.** Experiment with replacing the Category field currently used in your visuals with your newly created **Category Hierarchy**.

26. Save your work as:
NorthwindModeling_Step08.pbix

Make sure to commit and push your changes to week-09!

Module 4

Introducing DAX

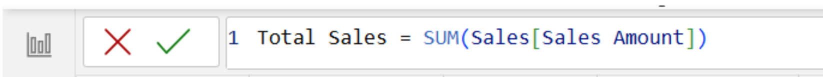
Exercise 4.A Northwind – Measures and Columns with DAX

1. Start by opening the Northwind modeling project as completed so far: **NorthwindModeling_Step08.pbix**

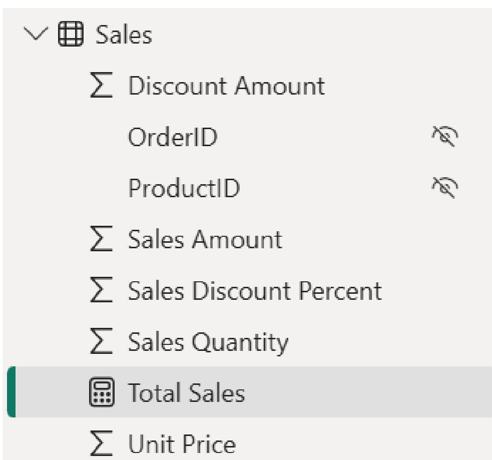
2. In the Table view, select the **Sales** table.

   * In the ribbon, go to the Table tools tab, and click **New measure**.

3. In the formula bar, create a measure named Total Sales by entering this formula:
   `Total Sales = SUM(Sales[Sales Amount])`



   * Click the green check (or hit enter) to save the measure. Notice that it now appears as one of the available fields on the Sales table. The calculator icon indicates that the field is a measure.



4. Head back to the Report view. In your report visualizations, try swapping out the existing Sales Amount values field in one or more visualizations with Total Sales:

* Notice that when you used Sales Amount in a visualization, the resulting label in the visual was “Sum of Sales Amount” -- but when you instead use the Total Sales measure, the resulting label just says “Total Sales.” This is because you are switching from an implicit measure created by Power BI to an explicit measure.

5. Back in the Table view, select the **Employee** table.

   * First, identify what information already exists in the table to form the basis of a calculated column for Employee Age. What column will you use to figure out ages?

6. In the **Table tools** tab of the ribbon, click **New column**.

   * Begin the formula with the name of your new column, `Employee Age =`

   * Start typing the name of the function `DATEDIFF()`, and hit tab or click to select when the function is suggested.

   * Note the structure of the function provided by the tooltip:
     `DATEDIFF(Date1, Date2, Interval)`
     where **`Date1`** represents the starting date, **`Date2`** the date against which to calculate the difference, and **`Interval`** meaning the unit in which the calculation should be returned (e.g. YEAR, MONTH, DAY, HOUR).

     - For Date1, use the column `Employee[Employee Birth Date]`

     - For Date2, use `TODAY()`

     - For Interval, use `YEAR`

   * The resulting complete expression should appear as follows:

     ```
     Employee Age = DATEDIFF(Employee[Employee Birth Date],
     TODAY(), YEAR)
     ```

   * When you hit enter or click on the green check to run the expression, you will see a new column appear at the end of the Employee table with the calculated current age for each employee:

Employee Age
57
46
42
39

Note: The Employee table should contain 9 total employees with no duplicates. If you notice that you have more rows than that, it may be that you experienced some duplication of records back when you originally added the Northwind database to MySQL Workbench. This is a natural checkpoint to notice this data quality issue. This issue may not occur, but if it does, go back to MySQL Workbench and use the DROP DATABASE option to delete Northwind, so that you have a clean slate. Then use the script from Week 2 to add Northwind again, and in Power BI, refresh your data. If you have any questions or issues, reach out to your instructor.

4. Using the same method and DATEDIFF() function, create another calculated column for **`Employee Years Employed`**, calculated using the existing column `Employee Hire Date` as the Date1 argument.

Employee Years Employed
7
7
7
6

5. Save your work as: **NorthwindModeling_Step09.pbix**

End of Week 9 Lab Workbook